

Validation of a Porosity-Turbulence Model in TrioCFD

Journée stagiaire | Ali Fadlallah

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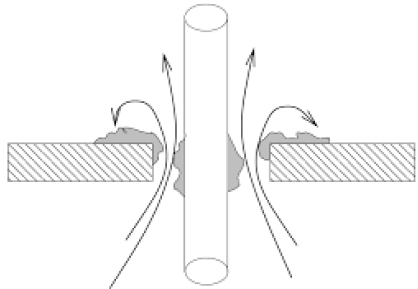


- 1** Introduction
- 2** The Porosity-Turbulence Model
- 3** Methodology
- 4** Results
- 5** Conclusion

1. Introduction:

- Context

- Clogging formation on spacer plates inside the steam generator.
- Coupling between:
 - **TrioCFD** (thermo-hydraulics).
 - **Scorpio** (chemical transport).
- The clogged region is modeled as a porous medium.
- New model in TrioCFD proposed as part of Antonin Couvez's PhD.



Clogging in a Steam Generator

1. Introduction:

- Objective



- **Validate** the implementation of the new porosity-turbulence model in **TrioCFD**.
- Compare it against 3 reference cases.

1. Introduction:

- Verification vs Validation



Verification

- Ensure the code solves the equations correctly.
- Focus on numerical accuracy.

Validation

- Ensure the model represents physical reality.
- Compare model predictions with reference or experimental data.

Our focus:

Validation of the porosity-turbulence model.

2. The Porosity-Turbulence Model:

- RANS



The Reynolds-averaged Navier–Stokes equations (RANS) can be written as:

$$\begin{cases} \nabla \cdot \phi \langle u \rangle_f = 0 \\ \partial_t \phi \langle u \rangle_f + \nabla \cdot \phi \langle u \rangle_f \langle u \rangle_f = -\frac{\nabla \langle p \rangle}{\rho} + \nu \nabla^2 \phi \langle u \rangle_f + \overline{g} - \nabla \cdot \langle u' u' \rangle + \phi \overrightarrow{P}_u + \delta_I \overrightarrow{Q}_u \end{cases}$$

- the flow quantities are decomposed into their time-averaged and fluctuating components.
- P terms: commutation and boundary terms.
- Q terms: interface jump between between porous medium and pure fluid.

2. The Porosity-Turbulence Model:

- Closing the system

Closure terms:

$$\vec{P}_u = \frac{\nu}{K} \phi \langle u \rangle_f + \frac{\alpha}{\sqrt{K}} \phi^2 |\langle u \rangle_f| \langle u \rangle_f$$

$$P_k = 0.28 \frac{\phi}{\sqrt{K}} \langle k \rangle_f \langle u \rangle_f$$

$$P_\epsilon = 0.56 \frac{\phi}{\sqrt{K}} \langle \epsilon \rangle_f \langle u \rangle_f + \phi^4 \frac{C_p^2}{60\kappa\nu} |\langle u \rangle_f|^4 \partial_t \beta$$

$$\vec{Q}_u = -\frac{C_p}{2} |\langle u \rangle_f| \langle u \rangle_f + (\nu_\phi + \nu) \frac{1-\phi}{6} \frac{L}{K} \left(\langle u \rangle_f \cdot \vec{t}_l \right) \vec{t}_l$$

$$Q_k = (\nu_\phi + \nu) \frac{1-\phi}{6} \frac{L}{K} \langle k \rangle_f$$

$$Q_\epsilon = (\nu_\phi + \nu) \frac{1-\phi}{6} \frac{L}{K} \langle k \rangle_f$$

$$C_p = c_0 \left(0.5(1-\phi) + \left(1 - \frac{1}{\phi} \right)^{2.2} \right)$$

$$K = 0.063 c_1 \frac{\phi^4}{1-\phi} L^2$$

$$\frac{\alpha}{\sqrt{K}} = \frac{0.774}{\sqrt{c_1}} \frac{1-\phi}{L \phi^{3.75}}$$



2. The Porosity-Turbulence Model:

- Important constants



$$C_p = c_0 \left(0.5(1 - \phi) + \left(1 - \frac{1}{\phi} \right)^{2.2} \right)$$

$$K = 0.063 c_1 \frac{\phi^4}{1 - \phi} L^2$$

$$\frac{\alpha}{\sqrt{K}} = \frac{0.774}{\sqrt{c_1}} \frac{1 - \phi}{L \phi^{3.75}}$$

- Two tunable constants which determine the behavior of the system: c_0 and c_1 .
- Fine-tune these constants to minimize the error.



- Implement the system of equations using Baltik in TrioCFD (already done by Antonin).
- Set up two configurations:
 - **Reference**: remove part of the mesh → solid obstacles.
 - **Validation**: porous region modeled by the new model.
- Use Jupyter Notebook to run and compare cases.
- Evaluate errors between validation and reference.

3. Methodology:

- Error Evaluation: L norms



- Compute error norms:
 - L_1 : mean relative error
 - L_2 : root mean square relative error
 - L_∞ : maximum relative error
- Compare velocity and pressure profiles
- For each combination of c_0, c_1 :
 - Run cases over porosity range ϕ : $0.1 \rightarrow 0.9$
 - Note: $\phi = 0$: fluid, $\phi = 1$: solid.
- Repeat for different c_0, c_1 combinations

3. Methodology:

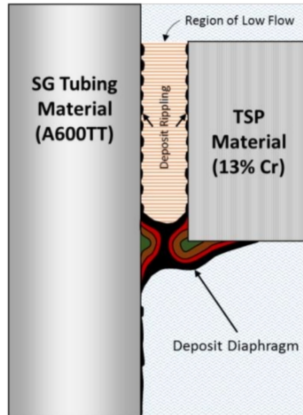
- Best Parameter Selection



- For each c_0, c_1 :
 - Compute L_1, L_2, L_∞ across all porosities.
 - Take mean of L_1, L_2 over ϕ .
 - Take max of L_∞ over ϕ .
- Identify the combinations that minimize:
 - Average L_1, L_2
 - Maximum L_∞
- Best (c_0, c_1) found: ...

4. Results:

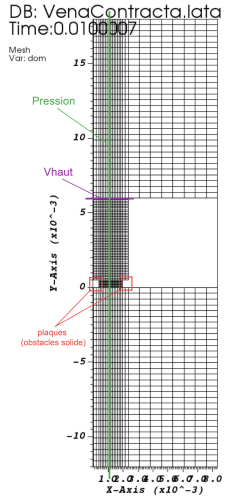
- Case 1



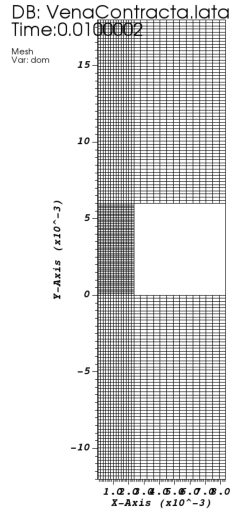
Experimental precipitation

4. Results:

- Mesh Layout



Reference case mesh



Validation case mesh

4. Results:

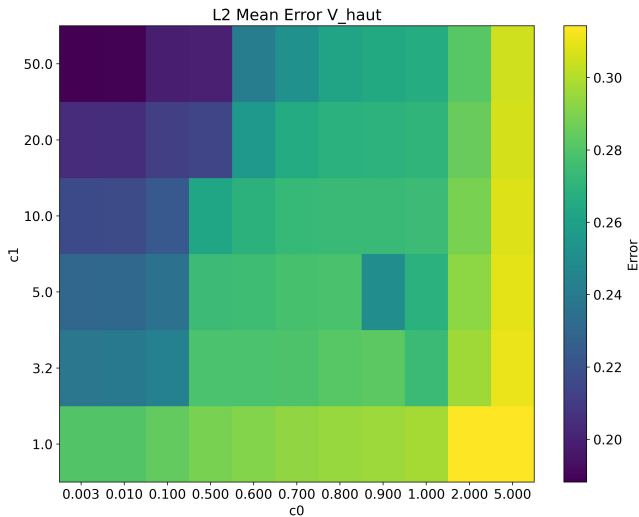
- Simulation



- Range tested: c_0 from 0.003 up to 5; c_1 from 1 up to 50.
- Porosity ϕ from 0.1 to 0.9.
- About 450 cases run in ~ 15 h.
- mesh size: ~ 2600 cells for the reference case; ~ 4000 cells for the validation case.

4. Results:

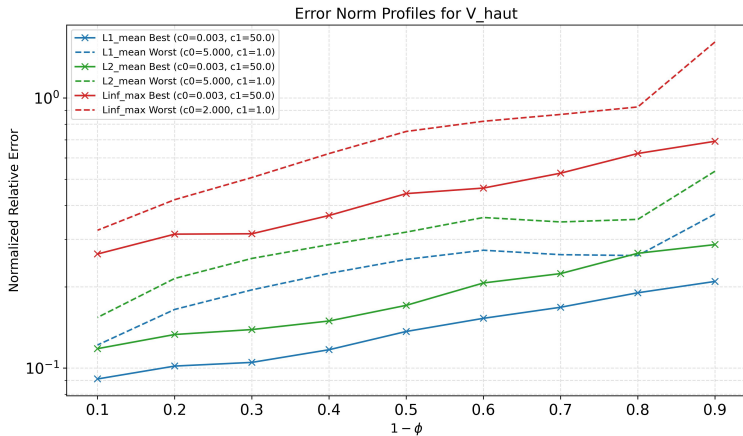
- Velocity error heatmap



L_2 error distribution vs c_0, c_1

4. Results:

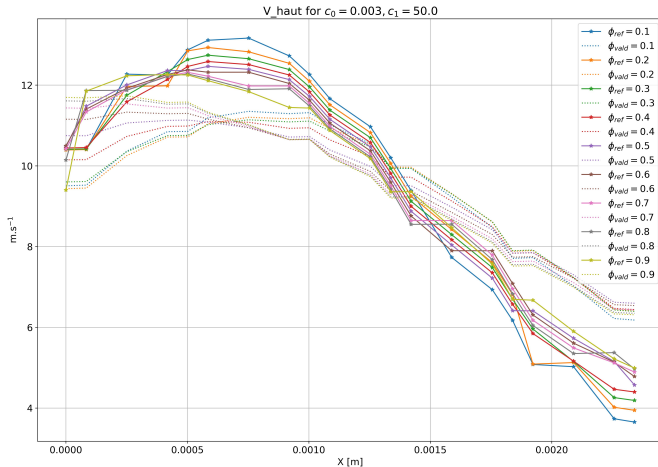
- Velocity error profiles



Best vs worst (c_0, c_1) combinations

4. Results:

- Velocity vs porosity



Velocity profile vs porosity for best (c_0, c_1)

4. Results:

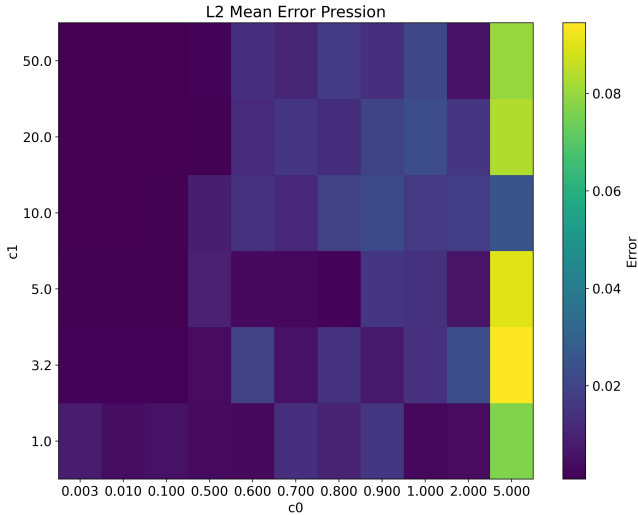
- Discussion (velocity)



- $c_0 = 0.003$, $c_1 = 50$ give lowest avg. L_1 , L_2 and min max L_∞
- Velocity errors remain high ($\sim 20\%$) due to turbulence complexity

4. Results:

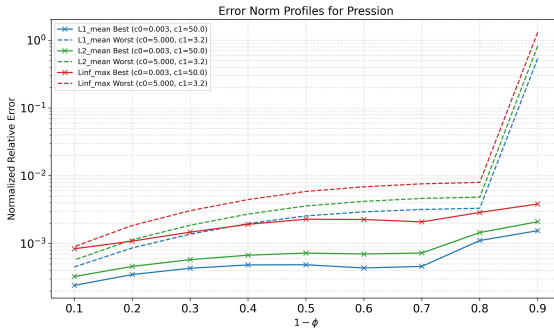
- Pressure error heatmap



L_2 error distribution vs c_0, c_1

4. Results:

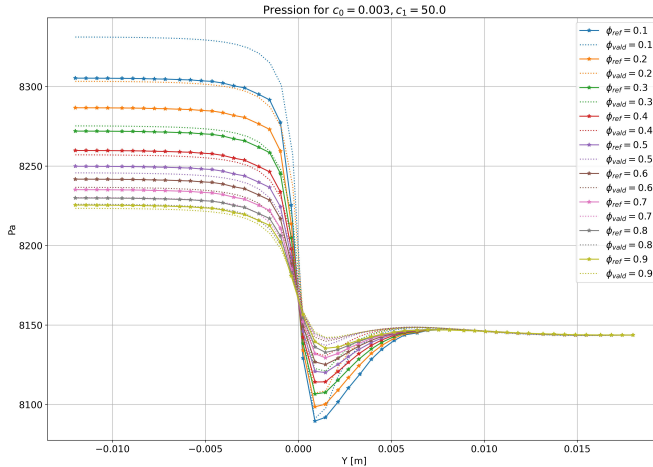
- Pressure error profiles



Best vs worst (c_0, c_1) for pressure

4. Results:

- Pressure vs porosity



Pressure profile vs porosity for best (c_0 , c_1)

4. Results:

- Discussion (pressure)



- Same $c_0 = 0.003$, $c_1 = 50$ also best for pressure.
- Pressure error much lower ($\sim 1\%$).
- Wider range of (c_0, c_1) still gives low pressure error.

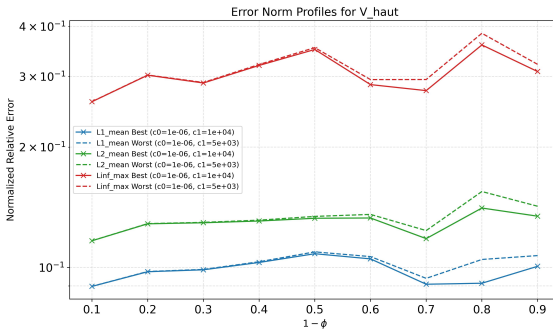
5. Conclusion



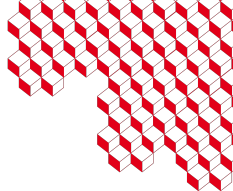
- Consistent best c_0 , c_1 found for both velocity and pressure.
- Observed trend: increasing c_1 reduces error but increases computation cost.
- Current work:
 - Extending c_0 , c_1 range (currently running!).
 - Validating on more test cases.

Appendix: Better Error ?

- Wider range of (c_0, c_1) tested.
- improved L_2 error to $\sim 12\%$ instead of $\sim 20\%$ for the velocity for $(c_0 = 10^{-6}, c_1 = 10^4)$.



Best vs worst (c_0, c_1) for velocity



Merci de votre attention

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